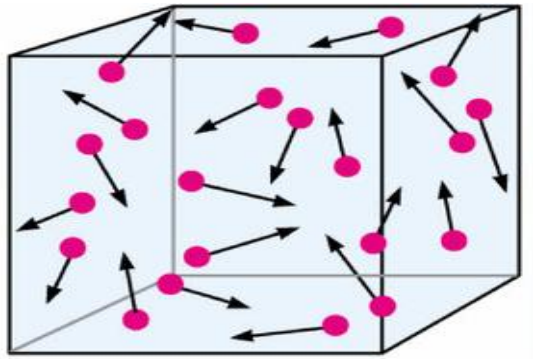
Fizika

9 sinf

Mavzu: Ideal gaz holatining tenglamalari



Ideal gazning holat tenglamasi

Ma’lum massali ideal gazning termodinamik holati uning uchta makroskopik parametrlari, ya’ni bosimi **p**, hajmi **V** va temperaturasi **T** orqali tavsiflanadi. Gaz bir holatdan ikkinchi holatga o‘tganida uning holatini tavsiflovchi (**p, V, T**) parametrlarning uchchalasi ham bir vaqtda o‘zgarishi mumkin. Gazning holat parametrlari (**p, V, T**)ning o‘zaro qanday bog‘langanligini ifodalovchi tenglama ***ideal gazning holat tenglamasi*** deyiladi.

Ideal gazning holat tenglamasini keltirib chiqarish uchun gazlar molekulyar-kinetik nazariyasining asosiy tenglamasidan foydalanamiz:

$p=nkT$ $\quad n = \frac{N}{V}$ $\quad N = \frac{m}{M} N_A$

$$pV = \frac{m}{M} N_A kT \qquad k \cdot N_A = R$$

$pV = \frac{m}{M} R\,T$ Mendeleyev- Klapeyron tenglamasi

Mendeleyev – Klapeyron tenglamasi

Mendeleyen-Klapeyron tenglamasi ideal gazning holatini aniqlagani uchun ideal gaz holatining tenglamasi deb ataladi.

$$pV = \frac{m}{M} RT$$

Ideal gazning holat tenglamasi gazning massasi, molyar massasi, bosimi, hajmi va temperaturasi orasidagi bog‘lanishni ifodalaydi.

Modda miqdori 1 mol bo‘lgan gaz uchun yozsak,

$pV = RT$ yoki $\frac{pV}{T} = R$

Klapeyron tenglamasi

Massa o‘zgarmas (m=const) bo‘lganda, gazning ikki holati uchun ideal gaz holat tenglamasini qo‘llaylik:

$$p_1V_1 = \frac{m}{M}\,R\,T_1 \qquad p_2V_2 = \frac{m}{M}\,R\,T_2$$

$$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$$

$\frac{pV}{T} = \text{const}$ – Klapeyron tenglamasi.

Gaz holatining uch parametridan bittasi noma’lum bo‘lib, qolgan ikkitasi ma’lum bo‘lganda, holat tenglamasi noma’lum parametrni aniqlashga imkon beradi.

$$\mathrm{p}V = \frac{m}{M}RT$$

$$\mathrm{T} = \frac{pVM}{mR} \qquad \mathrm{p} = \frac{mRT}{MV} \qquad \mathrm{V} = \frac{mRT}{Mp}$$

7-mashq

1-masala

Bosimi 0,45 MPa va temperaturasi 52 ℃ bo‘lganda 500 mol gaz qanday hajmni egallaydi?

Berilgan:

P=0,45 MPa=0,45·10^6 Pa

t = 52ºC

T= 52ºC+273= 325 K

ν = 500 mol

R = 8,31 J/K·mol

Formulasi:

$$P\,V = \frac{m}{M}\,R\,T \qquad \nu = \frac{m}{M}$$

$$P\,V = \nu\,R\,T$$

$$V = \frac{\nu\,R\,T}{P}$$

Topish kerak : V–?

Yechish: $$V = \frac{500\ mol \cdot 8{,}31\ \frac{J}{K \cdot mol} \cdot 325\ K}{0{,}45 \cdot 10^6\ Pa} = 3\ m^3$$

Javob: $\boldsymbol{V = 3\ m^3}$

7 – mashq

5 – masala

24ℓ hajmli ballonda 1,2 kg karbonat angidrid gazi bor. Ballon $3 \cdot 10^6$ Pa bosimgacha chidaydi. Qanday temperaturada portlash xavfi tug‘iladi?

Berilgan: CO_2

V=24l = 0,024 m^3

m = 1,2 kg

P =3·10^6 Pa

R =8,31 J/K·mol

M = 0,044 kg/mol

Topish kerak: t–?

Formulasi:

$$P\,V = \frac{m}{M}\,R\,T$$

$$\mathrm{T} = \frac{p\,V\,M}{m\,R}$$

$$t = T - 273$$

Yechish:

$$\mathrm{T} = \frac{3 \cdot 10^6 Pa \cdot 0{,}024\ m^3 \cdot 0{,}044\ kg/mol}{1{,}2\ kg \cdot 8{,}31\ J/K \cdot mol} \approx 318\ \mathrm{K}$$

$$\mathrm{t} = 318\ \mathrm{K} - 273 = 45^0\mathrm{C}$$

Javob: t = 45ºC

Mustaqil bajarish uchun topshiriqlar

1. Qanday tenglamaga ideal gazning holat tenglamasi deyiladi?
2. Mendeleyev – Klapeyron tenglamasini keltirib chiqaring.
3. Klapeyron tenglamasini va uning fizik ma’nosini yozing.
4. 7-mashqning 2,4,6,8-masalalarini yechish (32-bet).